# Software Requirements Specification

# for

# SPMS-Smart Parking Management System

**Version 1.0**

**Prepared by**
**A Sai Keerthan,424103**
**A Aditya,424102**
**G Anumitha Reddy,424129**

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# Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| SPMS | 07/08/2026 | Initial Version | 1.0 |

# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements of the Smart Parking Management System, a web-based application developed to simplify parking management. The system enables users to search for available parking spaces, reserve slots, make secure online payments, and access parking areas using QR code-based authentication. It also provides administrators with tools to manage users, parking lots, bookings, payments, and system analytics. The primary objective of this document is to define the complete functionality, performance requirements, constraints, and interfaces of the system, serving as a comprehensive guide for developers, testers, project managers, and other stakeholders throughout the software development life cycle.

## 1.2 Intended Audience

This document is intended for the users and stakeholders of the Smart Parking Management System, including parking lot administrators, and parking management personnel who will use, operate, and maintain the system. It is also intended for system administrators responsible for managing parking spaces, user accounts, bookings, payments, QR code operations, and overall system monitoring.

In addition, this document serves as a reference for developers, testers, project supervisors, project managers, and other stakeholders involved in the analysis, design, implementation, testing, deployment, and maintenance of the system. It provides a common understanding of the system requirements and ensures effective communication and coordination among all parties throughout the software development life cycle.

## 1.3 Intended Use

This Software Requirements Specification (SRS) document is intended to serve as the primary reference for the analysis, design, development, testing, deployment, and maintenance of the Smart Parking Management System. It defines the functional and non-functional requirements of the system and provides a common understanding of the expected behaviour, features, and constraints.

The document will be used by developers during implementation, testers for system verification and validation, project managers for planning and monitoring project progress, and stakeholders to ensure that the developed system satisfies the specified requirements. It also serves as a baseline for future enhancements, maintenance, and system upgrades.

## 1.4 Product Scope

The Smart Parking Management System is intended to manage parking-related activities through a centralized web-based platform. The system covers user registration, parking lot and slot management, slot availability, pre-booking and live booking, QR-based entry and exit, parking duration tracking, and payment processing.

The system also provides **administrative functions** for managing users, parking lots, slots, pricing, and reported parking issues. It maintains relevant **booking, parking session, and payment records** and provides users and administrators with appropriate information based on their access.

## 1.5 Definitions and acronyms

| Term | Full Description |
|---|---|
| Admin | Authorized person responsible for managing users, parking lots, bookings, and overall system operations |
| User | A registered individual who uses the system to search, reserve, and manage parking spaces |
| Parking Lot | A designated area containing multiple parking spaces available for vehicles |
| Parking Slot | An individual parking space assigned to a vehicle during a booking |
| Booking | Reservation of a parking slot for a specified duration |
| Live Booking | Booking a parking slot immediately based on current availability |
| Pre-Booking | Reserving a parking slot before the intended arrival time |
| QR Code | A Quick Response code generated for secure vehicle entry and exit |
| Vehicle | A two-wheeler or four-wheeler registered by a user for parking |
| Payment | The process of paying parking charges using supported online payment methods |
| Incident Report | A report submitted by a user regarding issues such as blocked slots, damaged slots, payment failures, or QR code problems |
| Dashboard | The main interface displaying parking information, bookings, and system status |
| DFD | Data Flow Diagram |
| EIS | External Interface Specifications |
| ERD | Entity Relationship Diagram |
| API | Application Programming Interface |

| | |
|---|---|
| SRS | Software Requirement Specifications |
| UI | User Interface |
| URS | User Requirement Specifications |
| SPMS | Smart Parking Management System |
| DBMS | Database Management System |

Table-1:Definitions and acronyms

## 1.6 Overview

- Section 2 – Overall Description: Describes the overall characteristics of the Smart Parking Management System, including the product perspective, product functions, user classes and characteristics, operating environment, design and implementation constraints, user documentation, and assumptions and dependencies.
- Section 3 – External Interface Requirements: Describes the interfaces through which users and external components interact with the system, including user, hardware, software, and communication interfaces.
- Section 4 – System Features: Describes the priority of the major system features and specifies the functional requirements of the Smart Parking Management System.
- Section 5 - Other Nonfunctional Requirements: Describes the nonfunctional requirements of the system, including performance, safety, security, software quality attributes, software backups and redundancy, and business rules.
- Section 6 – Summary: Provides a brief summary of the Software Requirements Specification document and the requirements described in the preceding sections.

# 2. Overall Description

## 2.1 Product Perspective

The Smart Parking Management System is based on the concept of efficient parking management. The primary stakeholders of the system are Users and Administrators. A user searches for available parking slots, reserves a suitable slot, completes the payment, and receives a QR code for parking entry and exit. The system maintains booking, payment, and parking records for future reference. Administrators manage parking lots, parking slots, users, bookings, payments, and incident reports while monitoring the overall parking operations through a centralized platform.

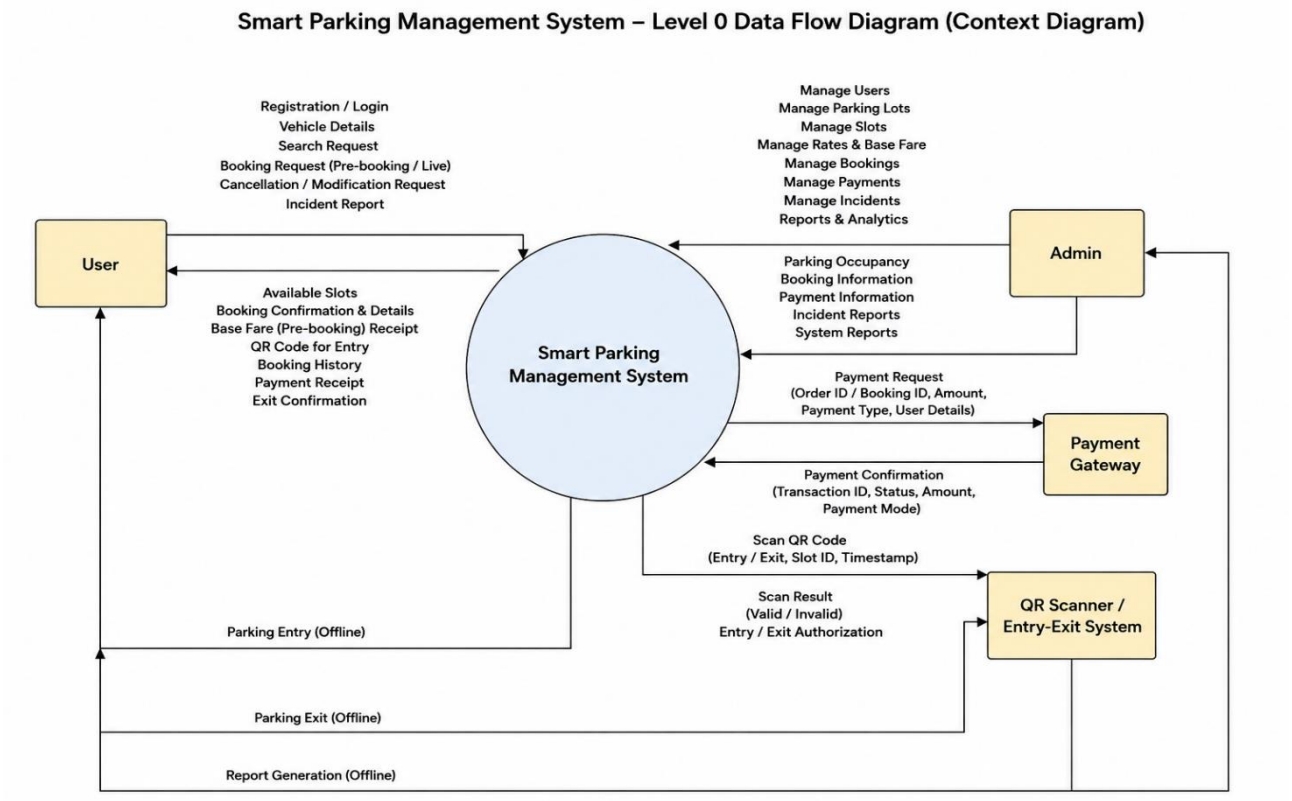



Figure 1-Context Diagram

## 2.2 Product Functions

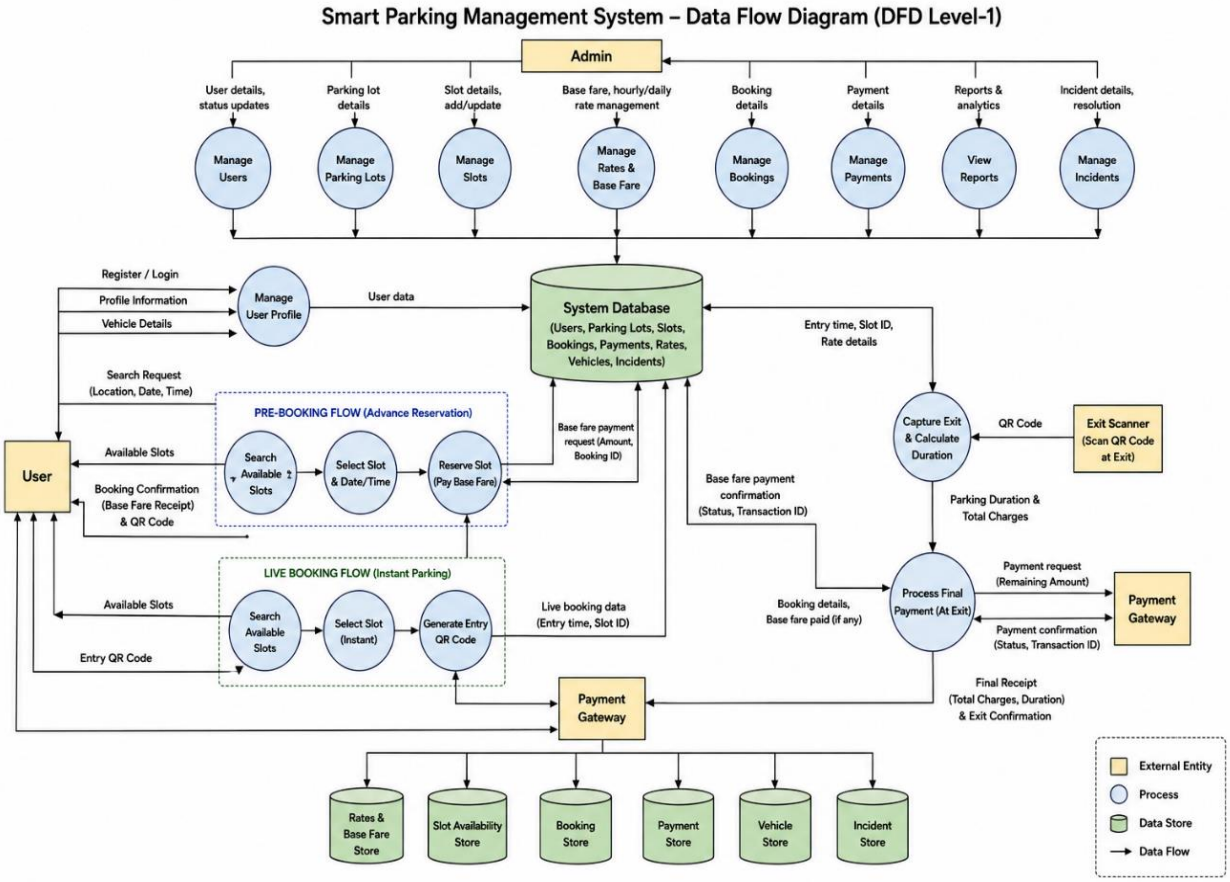



Figure 2-DFD

The above DFD clearly shows various processes associated with the project and how data flows between an entity and database via a specific process

The Admin manages the parking lots, parking slots, pricing, and user information. The admin also updates slot availability, manages bookings, resolves incident reports, and maintains the parking records required for the smooth functioning of the system.

A User registers into the system by providing personal and vehicle details. The user searches for available parking slots and can choose either Pre-Booking or Live Booking. For Pre-Booking, the user selects a slot, date, and time, pays the required base fare, and receives a booking confirmation along with a QR code. For Live Booking, the user selects an available slot and receives a QR code for immediate parking entry.

At the parking facility, the QR code is scanned to authorize entry. During exit, the system records the parking duration and calculates the total parking charge. If the booking is a Pre-Booking, the remaining parking amount is collected after deducting the base fare already paid. If it is a Live Booking, the complete parking charge is collected at exit. After successful payment, the system updates the booking and payment records and generates a digital receipt for the user.

**Non - Functional Features**:

The system should be able to scale to 10,000+ users concurrently and shall have a response time of ≤ 2 seconds for critical functionality. For other functions, the response time shall be ≤ 4 seconds during peak periods. Further, the system shall maintain consistent and accurate information, particularly for parking slot availability, bookings, parking duration, and payment details, ensuring that the information displayed to users reflects the actual system status.

## 2.3 User Classes and Characteristics

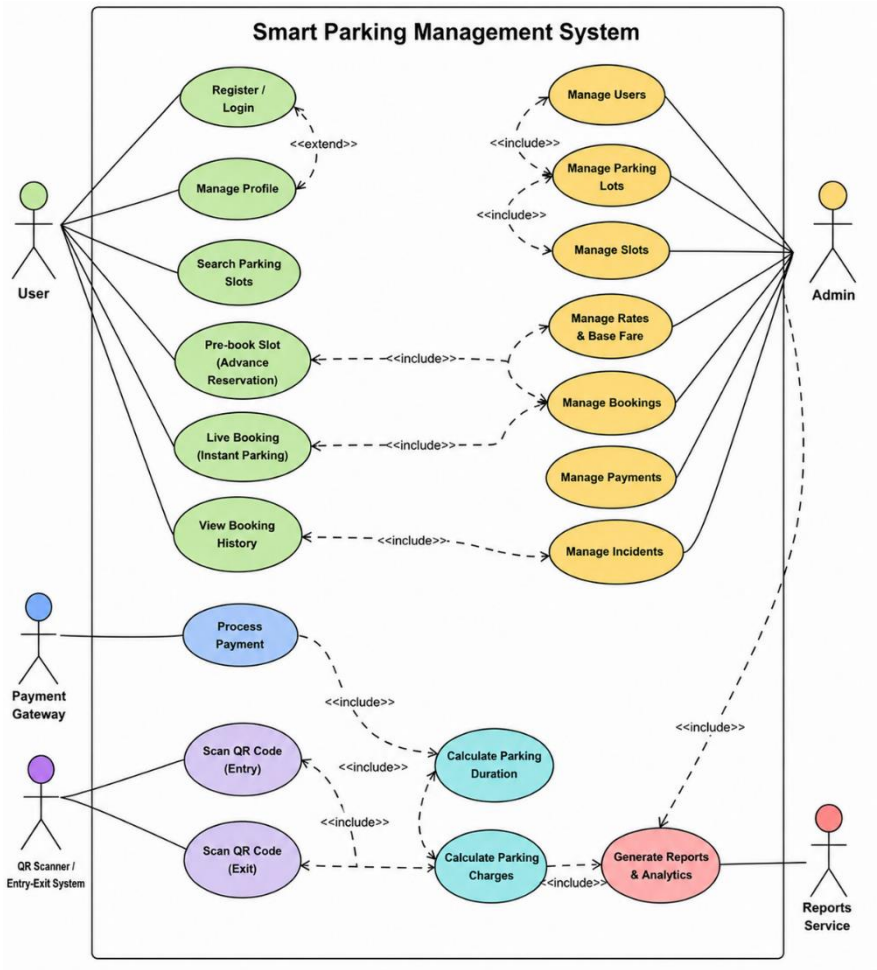



Figure 3-Use-case diagram

The actors and use cases are clearly shown in above USE-CASE diagram. The following are the Stakeholders/ Actors who perform the functions/ use cases as stated above:

Table 2-Use cases

| S. No | Actor Name | Description/ Actor's Role |
|---|---|---|
| 01 | Admin | ● Manage Parking Lots<br>● Manage Users<br>● Manage Parking Slots<br>● Manage Rates and Base Fare<br>● Manage Bookings<br>● Manage Payments<br>● Manage Incident Reports<br>● Generate Reports and Analytics |
| 02 | User | ● Register / Login<br>● Manage Profile and Vehicle Details<br>● Search Availability Parking Slots<br>● Pre-Book a Parking Slot<br>● Make Live Booking<br>● Scan QR code for Entry/Exit<br>● Make Payment<br>● View Booking History and Receipts<br>● Raise Incident Report |
| 03 | Payment Gateway | ● Receive Payment Requests<br>● Process Payments<br>● Provide Payment Confirmation and Transaction Details |
| 04 | QR Scanner / Entry-Exit System | ● Scan QR Code at Entry and Exit<br>● Validate QR Code<br>● Record Entry and Exit Time<br>● Send Entry / Exit Information to the System |
| 05 | Chemist | ● Generate Parking Reports<br>● Provide Analytics on Parking Occupancy, Bookings, Payments, and Incidents |

The following section describes the Use Cases with Pre-and Post Conditions:

Table 3: Pre-Post Conditions of use cases

| S. No | Use Case Name | Description | Pre-Condition | Post-Condition |
|---|---|---|---|---|
| UC-01 | User Registration | Register a new user with personal and vehicle details. | User is not registered and the registration information is available. | User account is created successfully. |
| UC-02 | User Login | Authenticate a registered user. | User has a registered account and provides valid credentials. | User is authenticated and can access the system. |
| UC-03 | Manage Profile | View or update personal and vehicle information. | User is logged into the system. | Updated information is saved successfully. |
| UC-04 | Search Parking Slot | Search for available parking slots. | User is logged in and parking information is available. | Available slots matching the search criteria are displayed. |
| UC-05 | Pre-Book Parking Slot | Reserve a parking slot in advance. | User is logged in and a suitable slot is available. | Slot is reserved, base fare is paid, booking is stored, and QR code is generated. |
| UC-06 | Live Booking | Book an available slot for immediate parking. | User is logged in and a parking slot is currently available. | Parking slot is assigned and a parking session is created. |
| UC-07 | Parking Entry | Validate the user's QR code and record entry. | User has a valid booking or parking session and QR code. | QR code is validated and entry time is recorded. |
| UC-08 | Parking Exit | Record exit and calculate parking duration. | User has an active parking session and is leaving the parking lot. | Exit time and duration are recorded and the final charge is calculated. |
| UC-09 | Process Payment | Process the parking payment through the payment gateway. | A valid payment request is generated. | Payment status and transaction details are recorded. |
| UC-10 | Pre-Booking Final Payment | Collect the remaining amount for a pre-booked slot at exit. | Pre-booking exists and parking duration is available. | Remaining amount is paid and the booking is completed. |
| UC-11 | Live Booking Final Payment | Collect the complete parking charge at exit for a live booking. | An active live parking session exists. | Total parking charge is paid and the parking session is completed. |

| S. No | Use Case Name | Description | Pre-Condition | Post-Condition |
|---|---|---|---|---|
| UC-12 | View Booking History | View previous and current booking records. | User is logged in and booking records exist. | Booking details and payment records are displayed. |
| UC-13 | Manage Parking Lot and Slots | Add, update, or remove parking lots and slots. | Admin is authenticated and authorized. | Parking lot and slot information is updated successfully. |
| UC-14 | Manage Users | Manage registered user accounts. | Admin is authenticated and authorized. | User information is added, updated, or managed successfully. |
| UC-15 | Manage Pricing | Configure parking rates and base fares. | Admin is authenticated and pricing configuration is available. | Updated pricing information is stored successfully. |
| UC-16 | Incident Reporting | Submit an incident related to parking. | User is logged in and has a parking-related issue. | Incident report is recorded for administrative action. |

## 2.4 Operating Environment

The Smart Parking Management System will be developed and operated using the following software, development tools, and environment:

Table 4-Environment for development

| Sr No | System | Environment for development | Description |
|---|---|---|---|
| 1. | **Frontend** | React | React is used to develop the user interface and provide an interactive web-based experience. |
| 2. | **Backend** | Node.js + Express | Node.js provides the server-side runtime and Express is used to build the backend services and APIs. |
| 3. | **Styling** | CSS, Tailwind CSS | CSS and Tailwind CSS are used to style the application and create a responsive user interface. |
| 4. | **Database** | PostgreSQL | PostgreSQL is used as the relational database for storing users, vehicles, parking lots, slots, bookings, payments, and related system information. |
| 5. | **Authentication** | JWT | JSON Web Token (JWT) is used for user authentication and secure access to protected system resources. |
| 6. | **Payment Gateway** | Razorpay | Razorpay is used to process online parking payments and handle payment transaction information. |
| 7. | **QR Code** | Node.js | Node.js is used to support QR code generation and related QR-based parking entry and exit operations. |
| 8. | **IDE / Code Editor** | Visual Studio Code | Visual Studio Code is used for writing, managing, debugging, and maintaining the application source code. |

| Sr No | System | Environment for development | Description |
|---|---|---|---|
| 9. | **Version Control** | Github | Github is used to track changes in the source code and manage different versions of the project. |
| 10. | **Operating System** | Windows 11 | Windows 11 is used as the primary development operating environment for developing and testing the system. |
| 11. | **Testing** | [TO BE UPDATED] | Testing environment and tools can be specified here based on the testing framework and tools used for the project. |

## 2.5 Design and Implementation Constraints

**The following Design and Implementation Constraints are applicable for the Smart Parking Management System:**

a) The system shall be designed to ensure that a parking slot cannot be assigned to more than one user for the same time period. The availability of slots must be checked before confirming a booking.
b) The system shall support both Pre-Booking and Live Booking. Pre-Booking requires the user to pay the applicable base fare during reservation, while the remaining parking amount shall be calculated and collected at the time of exit. For Live Booking, the parking charge shall be calculated based on the actual parking duration and collected at exit.
c) The system shall use QR code-based entry and exit validation. A valid QR code shall be associated with the corresponding booking or parking session, and the system shall record the entry and exit time to determine the parking duration.
d) The system shall maintain accurate records of users, vehicles, parking lots, parking slots, bookings, parking duration, and payment transactions. Access to administrative information shall be restricted to authorized administrators.
e) The system shall be designed to handle multiple users accessing parking availability and booking functionality at the same time without creating duplicate slot allocations or inconsistent booking information.
f) The system shall provide a responsive user interface so that users can access the system through commonly used desktop and mobile web browsers. The interface should remain usable across different screen sizes.
g) The system shall use secure authentication and authorization mechanisms to protect user and administrative information. Payment-related operations shall be handled through the selected payment gateway, and sensitive payment information shall not be directly stored by the system unless required.
h) The system shall be designed with a simple and intuitive interface so that users can easily search for parking slots, make bookings, complete payments, and perform entry and exit operations without unnecessary complexity.

## 2.6 User Documentation

The following documents shall be prepared:

1. Installation Guide
2. User Manual for end users

## 2.7 Assumptions and Dependencies

Key Assumptions are:

- The system shall assume that parking lots and parking slots have been registered and configured by the Admin before users can make bookings.
- Users shall be required to register and log in to the system before making a parking reservation or accessing their booking and payment information.
- The system shall support two types of parking: Pre-Booking and Live Booking. Pre-Booking allows a user to reserve a slot in advance, while Live Booking is intended for users who require parking immediately.
- For Pre-Booking, the user shall pay the applicable base fare at the time of reservation. The remaining parking amount, based on the actual parking duration, shall be calculated and paid at the time of exit.
- For Live Booking, the parking charge shall be calculated based on the actual parking duration and shall be paid by the user during the exit process.
- A parking slot shall not be assigned to multiple active bookings for the same applicable period. The system shall check slot availability before confirming a booking.
- QR codes shall be used for parking entry and exit validation. The QR code shall be associated with the relevant booking or parking session and shall be validated before recording entry or exit.
- The system shall record the entry time and exit time of a parking session. The parking duration shall be determined from these timestamps and used for calculating the applicable parking charge.
- Online payments shall be processed through the integrated payment gateway. The system shall record relevant payment details and payment status required for completing the booking or exit transaction.
- The Admin shall be responsible for maintaining parking lot information, parking slots, pricing details, and other administrative information required for operating the system.
- The system shall assume that users provide valid personal and vehicle information during registration and that the information required for booking and payment is available when needed.

- The system shall be accessible through supported web browsers and shall be designed to work on both desktop and mobile-sized screens.
- The system shall depend on the availability of the backend server, PostgreSQL database, payment gateway, and other external services required for the corresponding functionality.
- The system shall maintain booking, parking, and payment records for future reference, subject to the storage and database constraints of the deployed environment.
- The system shall not be responsible for physically controlling parking barriers, gates, or sensors unless such hardware integration is explicitly included in the project. QR validation and parking records shall be handled by the software system.

# 3. External Interface Requirements

## 3.1 User Interfaces

The User Interface section defines how users and administrators interact with the Smart Parking Management System. All screens shall be designed to work on mobile and laptop systems as well as different screen sizes through responsive design. Clear and descriptive error messages shall be displayed whenever an invalid input or unsuccessful operation occurs. Buttons and simple navigation elements shall be used to make the system easy to navigate. The parking search, pre-booking, live booking, payment, and entry/exit screens shall provide clear instructions and relevant information to help users complete the required operations easily.

Table 5-User Interfaces

| User No | User | User Interface Name | User Interface Description |
|---|---|---|---|
| 1 | All | Login / Registration | Allows users and administrators to register where applicable and securely log in to access the system. |
| 2 | User | User Dashboard | Displays parking availability, active bookings, parking sessions, and relevant user actions. |
| 3 | User | Profile Management | Allows users to view and update personal and vehicle information. |
| 4 | User | Search Parking | Allows users to search parking lots and view available slots based on required date, time, and location. |
| 5 | User | Pre-Booking | Allows users to select an available slot, pay the applicable base fare, and receive booking confirmation. |

| User No | User | User Interface Name | User Interface Description |
|---|---|---|---|
| 6 | User | Live Booking | Allows users to select an available slot for immediate parking and create a live parking session. |
| 7 | User | QR Code / Entry | Displays the QR code associated with a booking or parking session for parking entry validation. |
| 8 | User | Parking Exit | Displays entry time, exit time, parking duration, and the amount payable at exit. |
| 9 | User | Payment | Allows users to complete applicable payments through the integrated payment gateway and view payment status. |
| 10 | User | Booking History | Allows users to view previous bookings, parking sessions, and payment details. |
| 11 | User | Report Issue | Allows users to report parking-related issues, discomfort, or problems encountered while using the parking facility. |
| 12 | Admin | Admin Dashboard | Provides administrative access to parking lots, slots, users, pricing, bookings, payments, and system information. |
| 13 | Admin | Parking Lot and Slot Management | Allows the administrator to add, update, or remove parking lots and parking slot information. |
| 14 | Admin | User Management | Allows the administrator to view registered user information. |
| 15 | Admin | Pricing Management | Allows the administrator to configure and update parking rates and applicable base fares. |
| 16 | Admin | Booking and Payment Records | Allows the administrator to view booking and payment information for monitoring and management. |
| 17 | Admin | Manage Reports | Allows administrators to view submitted user reports, manage reported issues, and take appropriate action to resolve them. |

## 3.2 Hardware Interfaces

All the Stakeholders are supposed to log-in into the SPMS website where there will be a specific URL to access the System.

Hardware Requirements for stakeholders:

- Pentium 4 processor or higher
- Approximately 100 MB of free space
- Minimum 1 GB RAM

Hardware Requirements for hosting:

- Minimum 1GB database space
- Minimum 2GB RAM

| Sr No | Hardware Interface | Description |
|---|---|---|
| 1 | User Device | Desktop, laptop, tablet, or smartphone with a supported web browser can be used to access the system. |
| 2 | Camera | A device camera may be used to scan QR codes for parking entry and exit validation. |
| 3 | Internet Connection | A stable network connection is required for communication with the backend server, database, and payment gateway. |
| 4 | Server / Hosting Environment | The deployed application requires a server or cloud environment capable of running the Node.js and Express backend and connecting to PostgreSQL. |

## 3.2 Software Interfaces

The system shall be developed and deployed using the following software requirements:
Software Requirements for Hosting:

- Node.js
- Express.js
- PostgreSQL

Software Requirements for Stakeholders:

- Browser ( Google Chrome, Mozilla Firefox, Safari etc.)
- Operating System supporting the above browsers.

Software Requirements for Payments:

- Razorpay Payment Gateway for securely processing Debit/Credit Card, UPI, and other supported online payments.

Software Requirements for Development:

- React
- CSS/Tailwind CSS
- JWT
- Node.js
- Visual Studio Code
- Git

## 3.3 Communications Interfaces

The Smart Parking Management System will be available through a web-based URL and will operate using standard web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. Users and administrators will access the system through a secure encrypted connection using HTTPS to protect user, booking, and payment-related information during communication.

# 4. System Features

## 4.1 System Feature Priority Matrix

The following section describes the System features and their Priorities:

Table 6-Priority Matrix

| Sr No | Feature | Priority |
|---|---|---|
| 1 | User Registration and Login | High |
| 2 | Manage User Profile | Low |
| 3 | Search Parking Lots | Medium |
| 4 | View Parking Slot Availability | High |
| 5 | Pre-Booking | High |
| 6 | Live Booking | High |
| 7 | QR Code Entry and Exit | High |
| 8 | Parking Session and Duration Management | High |
| 9 | Payment Processing | High |
| 10 | Booking and Payment History | Medium |
| 11 | Report Parking Issue | Medium |
| 12 | Manage Parking Lots and Slots | High |
| 13 | Manage Users | High |
| 14 | Manage Pricing | High |
| 15 | Manage User Reports | Medium |
| 16 | Reports and Analytics | Medium |

## 4.2 Functional Requirements

The following section describes the Functional Requirements - Each Requirement has been tagged by a Requirement Number (Convention Used is Req-x and Req-x-x)

Table 7-Functional Requirements

| Requirement No | Functional Feature | Remarks |
|---|---|---|
| Req-1 | **User Registration and Login** | The system shall allow users to register and securely log in using their credentials. |
| Req-1-1 | **Register User** | The system shall allow a new user to provide the required personal and vehicle details to create an account. |
| Req-1-2 | **User Login** | The system shall authenticate registered users and provide access to the appropriate user functions. |
| Req-2 | **Manage User Profile** | The system shall allow users to view and update their profile and vehicle information. |
| Req-3 | **Search Parking Lots** | The system shall allow users to search for available parking lots based on the required location and parking requirements. |
| Req-4 | **View Slot Availability** | The system shall display the availability of parking slots in the selected parking lot. |
| Req-5 | **Pre-Booking** | The system shall allow users to reserve an available parking slot for a selected date and time. |
| Req-5-1 | **Reserve Slot** | The system shall reserve the selected slot after validating its availability and recording the booking details. |
| Req-5-2 | **Base Fare Payment** | The system shall allow the user to pay the applicable base fare required to confirm a pre-booking. |
| Req-5-3 | **Remaining Payment** | The system shall calculate the remaining parking amount based on the actual parking duration and allow the user to pay the remaining amount at exit. |
| Req-6 | **Live Booking** | The system shall allow users to create a live parking booking for immediate use when a suitable slot is available. |
| Req-6-1 | **Live Parking Session** | The system shall record the entry time and start the parking |

| | | |
|---|---|---|
| | | session for a live booking. |
| Req-7 | **QR Code Entry and Exit** | The system shall generate or provide a QR code for the applicable booking and use it to validate parking entry and exit. |
| Req-8 | **Parking Duration** | The system shall record entry and exit times and calculate the parking duration for the parking session. |
| Req-9 | **Payment Processing** | The system shall send payment requests to the integrated payment gateway and record payment confirmation and transaction details. |
| Req-9-1 | **Payment Confirmation** | The system shall update the booking or parking session after receiving successful payment confirmation from the payment gateway. |
| Req-10 | **Booking and Payment History** | The system shall allow users to view previous bookings, parking sessions, duration, and payment details. |
| Req-11 | **Report Parking Issue** | The system shall allow users to report parking-related issues, discomfort, or problems encountered while using the parking facility. |
| Req-12 | **Manage Parking Lots and Slots** | The system shall allow administrators to add, update, remove, and manage parking lots and their available slots. |
| Req-13 | **Manage Users** | The system shall allow administrators to view and manage registered user information. |
| Req-14 | **Manage Pricing** | The system shall allow administrators to configure and update parking charges and applicable base fares. |
| Req-15 | **Manage User Reports** | The system shall allow administrators to view submitted reports, manage reported issues, and record appropriate actions taken to resolve them. |
| Req-16 | **Reports and Analytics** | The system shall allow administrators to view relevant parking, booking, payment, slot occupancy, and parking-usage reports. |

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

At the peak, system should be able to scale to 10,000+ users concurrently. The following processes are critical and should respond within the specified time under normal operating conditions:

- **User Registration/Login** ≤ 3 seconds
- **Parking Lot and Slot Search** ≤ 3 seconds
- **Slot Availability Check** ≤ 3 seconds
- **Pre-Booking Request** ≤ 4 seconds
- **Live Booking Request** ≤ 4 seconds
- **QR Code Entry/Exit Validation** ≤ 2 seconds
- **Payment Request/Confirmation** ≤ 5 seconds
- **Booking and Payment History** ≤ 4 seconds
- **Report Issue Submission** ≤ 3 seconds

The system shall maintain consistent and accurate information during normal operation. The parking slot availability displayed to the user shall reflect the current status of the parking lot, including whether a slot is available, occupied, or reserved. Booking, parking duration, and payment details shall be updated correctly to ensure that users and administrators always receive reliable information.

## 5.2 Safety Requirements

The system shall be designed to ensure the **safe and reliable operation of the parking facility**. The parking lot layout, parking slots, entry and exit points displayed in the system shall accurately represent their actual physical locations. The system shall prevent incorrect or conflicting slot assignments that could lead to vehicle collisions or unsafe parking conditions.

The system shall ensure that **parking slot information cannot be manipulated or altered by unauthorized users**. Any changes to parking lot layouts, slot information, or availability shall be performed only by authorized administrators. The system shall also ensure that faulty or unavailable parking slots(as marked by administrator) are not shown as available for booking.

## 5.3 Security Requirements

The system shall maintain the confidentiality of user, booking, and payment-related information. Users shall be provided access only to their own personal and parking-related information, while administrators shall be permitted to view only the basic information required for managing the parking system. Authentication and authorization shall be implemented to prevent unauthorized access to the system. Access to sensitive information and system functions shall be restricted based on the user's role, and unauthorized users shall not be allowed to modify booking, parking, or payment information.

## 5.4 Software Quality Attributes

The Key Software Quality Attributes are **Reliability, Usability, Security, and Maintainability**. Since the system handles parking bookings, slot availability, and payment transactions, it needs to be **highly reliable and consistent**. The system should provide a simple and user-friendly interface so that users can easily search, book, make payments, and complete parking entry and exit operations. **Security** is important to protect user and payment-related information from unauthorized access. The system should also be **maintainable**, allowing administrators and developers to manage parking information and make necessary system updates efficiently.

## 5.5 Software Back-ups and Redundancy

The system data shall be backed up **every night** with a retention cycle of **30 days**. The backup shall include user, parking, booking, parking session, and payment-related data to provide a provision for recovery in case of data loss or system failure.

The backup shall be periodically restored in a **separate test environment** to verify the completeness and correctness of the backup. The test data shall be removed after successful verification. A redundant copy of the backup shall also be maintained separately from the primary system to reduce the risk of data loss due to system failure.

# 6. Summary

This document details the **Software Requirements Specification (SRS)** for the **Smart Parking Management System**. The document has **5 sections**. The first section, **Introduction**, describes the purpose of the document, intended audience, intended use, product scope, definitions and acronyms, and an overview of the system.

**Section 2, Overall Description**, provides a detailed description of the system. It covers the product perspective, product functions, user classes and characteristics, operating environment, design and implementation constraints, user documentation, and assumptions and dependencies.

**Section 3, External Interface Requirements**, describes the interfaces of the system, including user interfaces, hardware interfaces, software interfaces, and communication interfaces.

**Section 4, System Features**, describes the system feature priority matrix and the functional requirements of the Smart Parking Management System. The functional requirements define the major operations and services provided by the system.

**Section 5, Other Nonfunctional Requirements**, describes the nonfunctional requirements of the system. It includes performance requirements, safety requirements, security requirements, software quality attributes, software backups and redundancy.